Causal Mediation Analysis Reveals Syntactic Agreement Mechanisms in Neural Language Models

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Abstract

Targeted syntactic evaluations have demonstrated the ability of language models to perform subject-verb agreement given difficult contexts. Although this is well established, the mechanisms by which neural language models achieve syntactic agreement are still not well understood. As a remedy, this thesis applies causal mediation analysis to pre-trained neural language models to locate model components and discover mechanisms responsible for predicting correctly inflected verbs. In particular, we investigate the magnitude of models' grammatical inflections preferences, as well as compare which neurons process subject-verb agreement across sentences with different syntactic structures. In our results, we uncover both similarities and differences across architectures and model sizes, and get a glimpse at the within-model mechanisms that produce number agreement. Notably, we learn that larger models do not necessarily learn stronger preferences, we observe two distinct mechanisms for producing subject-verb agreement depending on the syntactic structure of the input sentence, and we find that language models rely on similar sets of neurons when given sentences with similar syntactic structure.

Acknowledgements

Introduction

This thesis is an adaptation of an paper anonymously submitted to the The Joint Conference of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP 2021). The paper was co-authored[1] by myself and Aaron Mueller[2], and co-advised[3] by Stuart Shieber[4], Sebastian Gehrmann[5], Tal Linzen[6], and Yonatan Belinkov[7].

The presentation of our work is split into two parts. In part I, we give background and motivation for our methods. Chapter 1 gives a brief introduction to neural language models (LMs) and introduces previous work in targeted syntactic evaluations, and probing studies of neural LMs. The latter half of the chapter is devoted to an overview of causal mediation analysis, defining terms and giving examples before outlining the formulation of causal mediation analysis for natural language processing, from which we derive the metrics we analyze later. Chapter 2 outlines the setup of our experiments including a description of our data, models, and experimental methods. Part II focuses on the contributions of this thesis. It gives a summary of our major results, and is split into chapters for different metrics we used to analyze the language models.

Targeted syntactic evaluation studies[8] have shown that neural language models (LMs)[9] are able to predict the correct token from a set of minimally differing continuations the vast majority of the time, even in difficult contexts [Linzen et al., 2016, Gulordava et al., 2018]. This is true for a number of linguistic tasks including subject-verb agreement [van Schijndel et al., 2019], filler-gap dependencies [Wilcox et al., 2018], and reflexive anaphora [Marvin and Linzen, 2018]. For example, van Schijndel et al. [2019] show that recurrent neural

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[6] New York University
[7] Technion University, Israel
[8] Introduced in section 1.2
[9] Neural language models, including LSTMs and Transformers are introduced in section 1.1.

networks are able to learn subject-verb agreement rules, even when challenged with an attractor, such as in the following example.

(1) The key to the cabinets is/*are next to the coins.

In this case, a model using a linear analysis (i.e., inflecting the verb based on the most recent noun) would favor the ungrammatical inflection, whereas a model using a hierarchical analysis would choose the grammatical inflection.

While we have a reasonable understanding of the generally correct *behavior* of LMs in such contexts, the mechanisms behind the models' sensitivity to syntactic agreement are still not well understood. Recent work has performed causal analyses units within LSTM models. For instance, Lakretz et al. [2019] uncover neurons responsible for syntactic agreement by observing the effects of ablating specific neurons, and Giulianelli et al. [2018] perform causal analyses of LSTM hidden representations' impact on syntactic agreement using "diagnositc classifiers" trained to to predict number from the internal states of the language model. While these studies focus on LSTM models, Transformer-based LMs' agreement mechanisms have not been investigated. Transformer-based LMs' are therefore enticing candidates for further analysis given their superior syntactic generalization abilities compared to those of LSTMs [Hu et al., 2020].

We thus propose applying the behavioral-structural method of causal mediation analysis as put forward by Pearl [2001] to syntactic agreement, following the approach used by Vig et al. [2020] for interpreting gender bias in pre-trained LMs. This method allows us to implicate specific model components in the observed behavior of a model. If we view a neural LM as a causal graph proceeding from inputs to outputs, we can view each model component (e.g., a neuron) as a mediator. We measure the contribution of a mediator to the observed output behavior by performing controlled *interventions* on input sentences and observing how they change the probabilities of continuation pairs. We focus primarily on GPT-2 [Radford et al., 2019], although we also analyze TransformerXL [Dai et al., 2019] and XLNet [Yang et al., 2020].

As a result of our analysis, We find that both GPT-2 and Transformer-XL use two distinct mechanisms to accomplish subject-verb agreement, one of which is active only when the subject and verb are adjacent. Conversely, XLNet shows more unified behavior across syntactic structures. We also find that even though larger models assign a higher probability to the correct inflection more often, this does not necessarily translate to a larger margin between the probability of the correct and incorrect options. Additionally, in larger models, agreement mechanisms are similar to those in smaller models, but more distributed across layers. Finally, we find that the most important neurons for agreement are shared across different structures to various extents, and that the degree of neuron overlap matches well with human intuitions of syntactic similarity between structures.

Part I

Preliminaries

Chapter 1

Background and related work

This chapter gives an overview of previous work that motivates and informs the experiments we perform in part II. We begin with an overview of language modeling and neural language models. We then briefly discuss previous work in syntactic analysis of language models (section 1.2) and probing (section 1.3), then give background for causal mediation analysis before introducing it as a tool for within-model syntactic evaluation in section 1.4.

1.1 Neural networks as language models

Statistical language models (LMs) are probability distributions over word sequences. Building language models that match the distribution of a human language is a major challenge that has gotten considerable attention because of LMs' immense usefulness for a number of natural language processing tasks. Training artificial neural networks as language models using machine learning methods have yielded impressive results that warrant investigation.

Our work focuses on a class of neural network language models called Transformers [Vaswani et al., 2017]. A Transformer is a function that takes a sentence (encoded as a vector) as an input, applies a sequence of linear and nonlinear transformations, and outputs a probability distribution over a vocabulary for the next word (again encoded as a vector). The transformations are parameterized by two sets of model components: neurons and attention heads. A Transformer is composed of alternating layers of neurons and attention heads, where the neurons greatly outnumber the attention heads, with hundreds of neurons and only a few attention heads per layer. The exact details of how these components work are beyond the scope of this thesis, but it suffices to understand that "weights", or settings of the neurons and attention heads determine the behavior of the

Transformer. Machine learning algorithms use massive amounts of training data to fine tune the neuron and attention head parameters to achieve state-of-the-art performance on language modeling tasks.

Another common neural language model architecture that has achieved impressive results in the past (before Transformers came to dominate) is the Long Short-Term Memory (LSTM) architecture. LSTM language models are recurrent neural networks that operate by maintaining "hidden" states that are updated at each time-step as the model ingests word sequences one token at a time. The sequential nature of LSTMs contrasts with the massively parallel computation method in Transformers. Parallel computation enables efficient training of Transformer language models and allows them to surpass LSTM models in language modeling performance. Again, we will not go into great detail on how LSTM models work, and a high-level understanding of these architectures is sufficient to grasp the experiments and results in this thesis.

One downside to neural language modeling is that it is not immediately apparent what these models learn that allows them to approximate the distribution of natural language so well. This problem gives rise to the phrase "black-box" in describing neural networks.

1.2 Targeted Syntactic Evaluation

As black-box neural language models have rapidly gained popularity and improved over the last decade, interest in evaluating these models' ability to perform non-trivial linguistic tasks has increased. Towards this goal, recent years have seen a number of behavioral studies of neural language models as psycholinguistic subjects [Linzen et al., 2016, Gulordava et al., 2018, Goldberg, 2019], focusing primarily on LSTM LMs [Sundermeyer et al., 2012], GPT-2 [Radford et al., 2019], and BERT [Devlin et al., 2019a][1] as subjects. Among the more popular linguistic phenomena to test are whether models prefer grammatical completions in subject-verb agreement contexts [Marvin and Linzen, 2018, van Schijndel et al., 2019, Goldberg, 2019, Mueller et al., 2020], as well as in filler-gap dependencies [Wilcox et al., 2018, 2019] because these tasks supposedly require a nontrivial knowledge of the hierarchical syntactic structure of language. These targeted syntactic evaluations are largely based on the approach of Linzen et al. [2016], where a model's ability to syntactically generalize is measured by its ability to choose the correct inflection in difficult structural contexts instantiated by tokens that the model has not seen together during training. The model's ability to choose the correct inflection is tested by checking whether the model assigns the correct inflection a higher probability than an incorrect inflection given the same context.

The targeted syntactic evaluation approach effectively investigates the output behavior of the model, but does not inform us of inner workings of the model or

[1] BERT is a contextualized word prediction model, not a true language model.

identify components responsible for the observed behavior. This thesis builds upon this basic approach by incorporating component-level analysis of the model's behavior.

1.3 Probing

A separate line of syntactic analysis work has attempted to look inside black-box neural language models by training classifiers to probe model component outputs. This approach investigates representations within the model associated with syntactic dependencies by defining a family of functions (probes) that map from model representations to some phenomenon that those representations are expected to encode. Hewitt and Manning [2019] and Chi et al. [2020] map BERT representations to dependency parses as a proxy for discovering latent syntactic knowledge within the model. More related, Giulianelli et al. [2018] use probes to investigate how LSTMs handle agreement.

Probing is more difficult to interpret than behavioral approaches because the addition of a trained classifier introduces confounds [Hewitt and Liang, 2019]: most notably, whether the probe maps from model representations to the desired output, or learns the task itself. Probes also only give correlational evidence, rather than causal evidence [Belinkov and Glass, 2019] for component implication in syntactic behavior. Importantly, when using probes, one does not observe behavior in syntactically controlled settings, instead evaluating on tasks such as dependency parsing/labeling; thus, unlike in behavioral studies, we cannot easily make observations about differing model behaviors for different syntactic structures. Our approach, introduced later, attempts to overcome the aforementioned shortcomings by causally implicating model components in syntactic behavior for controlled linguistic structures.

1.4 Causal Mediation Analysis

To accomplish our goals, we turn to the methods of causal mediation analysis. Causal inference methods study the change in a response variable following an intervention; for example, how do health outcomes change after a patient stops consuming nicotine products? Causal mediation analysis [Robins and Greenland, 1992, Pearl, 2001, Robins, 2003] focuses on the role of a mediator in explaining the effect of a treatment on outcomes. For example, if a patient stops using tobacco, are health outcomes mediated by the initial method of nicotine delivery (e.g., smoking tobacco vs. patches vs. nicotine gum)?

This approach lends itself well to interpreting NLP models, as we can view a deep neural network as a graphical model from input to output via mediators, where mediators can be individual components (e.g., neurons). For LMs, the intervention is a change to the input sentence, and the outcome is a function of the probabilities of a set of continuations.

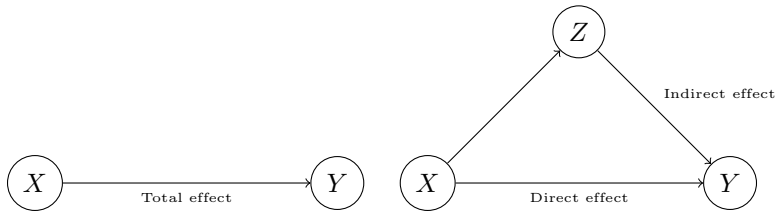

Figure 1.1: Total effect vs. direct effect vs. indirect effect.

Section 1.4.1 provides an overview of the terms and definitions (an illustration of which can be found in fig. 1.1) used in causal mediation analysis. With this framework, we eventually pursue an interpretation of the mechanisms producing grammatical predictions within neural language models.

1.4.1 Total, direct, and indirect effects

Causal inference methods attempt to infer the causal relationships between variables. These methods have been developed extensively for use in many fields including economics [Maziarz, 2020] and social science [Mahoney, 2010]. The ability to conclusively link the causal effect of one variable on another is particularly valuable in the fields like medicine [Hill, 1965] where the researchers may wish to compare the effects of interventions such as surgery and drugs on patient outcomes. In this section, we use a modified version of the drug treatment scenario from Pearl [2001] as an illustrative example to motivate our own formulation of causal mediation analysis for neural language models. In our example, doctors wish to evaluate the causal effects of a drug treatment intended to treat a disease.

Total effect

In investigating the causal relationship between an independent variable X and dependent variable Y, it makes sense to measure the value of Y as a function of X. This straightforward relationship is known in causal inference literature as *total effect*.

As an example, to evaluate the effectiveness of a drug treatment, medical doctors might design a randomized experiment where some proportion of disease patients receive the treatment and the remaining patients receive no treatment. After administering (on not administering) the treatment for a period of time, doctors could measure the effectiveness of the drug by calculating difference in disease rates among those who did and did not receive treatment. If the difference is large, then it can be concluded that the drug is effective. To formalize this, let $X = x$ be the case where the patient receives the treatment, $X = 0$ be the case where the patient receives no treatment, $Y = y_x$ be disease rate for patients receiving the treatment, and $Y = y_0$ be the disease rate for patients receiving no

treatment. Doctors may then be interested in the proportional difference

$$\frac{y_x - y_0}{y_0} \tag{1.1}$$

as a metric for the drug's total effect.

Direct effect

Although useful, total effects can be misleading. In our disease treatment example, suppose the drug has a side effect of producing headaches. As a result, patients put on the drug end up consuming more aspirin than those in the control group. If it turns out that aspirin is an effective treatment for the disease, the patients under drug treatment end up with lower disease rates than the control group, despite the possibility the drug itself has no *direct effect* at all. It is therefore important for doctors to be able to identify direct effects of a drug treatment.

In this case, in addition to the independent variable X (treatment) and the dependent variable Y (disease rate), we must contend with a *mediating variable* Z. To find the direct effect of X on Y, controlling for Z, we can measure the proportional difference in response Y under treatments $X = x$ and $X = x^*$, while holding $Z = z$ constant.

$$\frac{y_{x^*z} - y_x}{y_{0z}} \tag{1.2}$$

where y_{xz} and y_{0z}

In order to find the direct effect of the drug treatment, doctors can control the mediating variable Z by holding aspirin intake constant for both experimental groups and calculate the direct effect as the (proportional) difference in disease rates for patients who receive and do not receive the drug treatment.

Indirect effects

Now that we now can find the direct effect of drug treatment on disease rates, we may now be interested in quantifying the mediating effects of other variables such as aspirin intake. In contrast to total and direct effects, which tell us whether a treatment has an effect, the *indirect* effects of the mediating variables allows us to understand *how* a treatment works.

To find the indirect effect of a variable Z which mediates the effect of independent variable X on Y, we can measure the proportional difference in the response $Y = y$ under the treatment $X = x$ while holding Z to the value z^* it takes when $X = x^*$ and the value z it takes when $X = x$. This gives us

$$\frac{y_{xz^*} - y_x}{y_x} \tag{1.3}$$

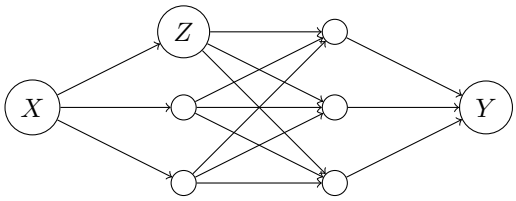

Figure 1.2: A neural network as a causal graph.

In our example, this is the (proportional) difference in disease rates among two groups of patients not given the treatment, where one group takes aspirin at a level that they would have had they been given the treatment, and the other group takes aspirin as they would usually.

1.4.2 Causal mediation analysis for neural networks

In part II, we adapt a methodology developed by Vig et al. [2020] for applying causal mediation analysis (as formulated in Pearl [2001]) to neural networks to study number agreement. In causal mediation analysis for NLP, we view a neural network as a causal graph proceeding from inputs (X), such as linguistic prompts, to output completions (Y), mediated by model components (Z) such as neurons (see fig. 1.2). This setup allows us to implicate specific model components in the behavior of the neural network.

In their work, Vig et al. [2020] implicate specific neurons and attention heads in mediating gender bias within various pre-trained LMs. In our work, we do the same for neurons and attention heads mediating syntactic number agreement. One main difference in our application is that, while one ideally expects equal preferences for male and female completions given gender-ambiguous contexts (for example, given the prompt u "The nurse said that", we expect similar continuation probabilities $p(\text{she}|u) \approx p(\text{he}|u)$), this is not the case for subject-verb agreement, where we expect very strong preferences for grammatically correct completions over incorrect completions. A similar assumption of strong preference is also used in recent work by Geiger et al. [2020] who use interventions as a tool to verify that a BERT-based [Devlin et al., 2019b] natural language inference model has learned a specific algorithm.

By applying causal mediation analysis to neural nets, we arrive at a novel method for syntactic evaluation of neural language models that affords us more fine-grained understanding of model behavior in controlled linguistic contexts than previous methods mentioned in section 1.2, while avoiding many of the drawbacks and limitations of the probing methods from section 1.3.

Chapter 2

Experimental setup

With the goal of discovering which model components are responsible for subject-verb agreement, we begin by constructing a data set of natural language examples where both short- and long-range syntactic dependencies exist between the subject and the verb. We also construct a set of examples where there are 'attracting' nouns that might cause confusion as to which noun the verb ought to agree with. After constructing these examples, we select a number of freely available pre-trained language models on which to perform our interventions, with the goal of gathering data to calculate the total, direct, and indirect effects of syntactic structures and model components on the model output.

2.1 Data

As our set of natural language examples, we define a set of prompts $\mathbf{u}$. These prompts are a set of left contexts (beginnings of sentences), generated using from a set of templates and vocabulary developed by Lakretz et al. [2019] and augmented to include more syntactic structures, such as relative clauses, and a few more vocabulary tokens, borrowed from Marvin and Linzen [2018]. We opt to synthetically generate prompts rather than sample from a corpus to control for the confound of token collocations in the training set. We use templates for six syntactic structures; an example of each may be found in fig. 2.1. For each structure, we randomly sample 300 prompts from the full set of prompts generable using our vocabulary. The vocabulary we use is given in appendix A. Our dataset, code, and random seeds are available on GitHub.[1]

In the 'simple agreement' and 'within RC' constructions, there is no separation between the target subject and verb. The 'across one distractor' and 'across two distractors' structures test the effect of placing an adverb/adverbs between the subject and verb. The 'across PP' and 'across RC' structures test the effect

[1] https://github.com/mattf1n/lm-intervention

Simple Agreement:
The athlete is/*are

Within Object Relative Clause:
The friend (that) the lawyers *likes/like

Across One Distractor:
The kids gently *admires/admire

Across Two Distractors:
The father openly and deliberately avoids/*avoid

Across Prepositional Phrase:
The mother behind the cars approves/*approve

Across Object Relative Clause:
The farmer (that) the parents love is/*are

Figure 2.1: Syntactic structures used in this study. Ungrammatical forms are marked with asterisks. Target subjects and their agreeing verb inflections are shown in blue, while attractors and their agreeing inflections are shown in red.

of adding a noun (and verb in the latter structure) between the main subject and the main verb. In the 'across RC' and 'within RC' structures, we measure effects both with and without the complementizer *that*.[2]

In each of these constructions, we define a correct and an incorrect continuation. Here, we focus on the third-person singular/plural distinction.

2.2 Models

In this thesis, we focus primarily on GPT-2 [Radford et al., 2019], an auto-regressive Transformer-based English LM. Auto-regressive means that during text generation, the model generates word sequences one token at a time, inputting the previous tokens back into itself in order to produce the next. We use several model sizes: small, medium, large, and extra-large. We also use DistilGPT-2 [Sanh et al., 2020], a very small distilled version.

To investigate how differences in training across Transformer-based architectures manifest themselves in syntactic agreement mechanisms, we also investigate Transformer-XL [Dai et al., 2019] and XLNet [Yang et al., 2020]. Transformer-XL is an auto-regressive English LM whose training objective is similar conceptually to GPT-2's; however, it has a much longer effective context due to its segment-level recurrence. XLNet is an English LM which proceeds through all

[2] A comparison of total and indirect effects when including or excluding the complementizer may be found in section 4.3.

permutations of the input tokens during training, and which uses a distinct attention masking mechanism as well; during testing, it proceeds through the input in an auto-regressive manner, similar to the other two models.

2.3 Experiment

For each model m, model component (neuron or attention head) Z, template U, noun n, and verb v, we create prompts $u \coloneqq U(n)$ and $u' \coloneqq U(n')$ where n and n' are the singular and plural subjects of the prompt, and generate the next-word predictions $m(u)$ and $m(u')$ which are a probability distributions over the vocabulary. We then record $\Pr(v)$ and $\Pr(v')$, the probabilities of the singular and plural inflections of the verb v from each distribution. Using z and z', the outputs of component Z under the u and u' prompts, we then record $\Pr(v \mid z')$ and $\Pr(v' \mid z)$, the probabilities of the singular and plural inflections of the verb v as given by the outputs $m(u, z')$ and $m(u', z)$, which are the model outputs when component Z is set to the second argument.

The probabilities recorded using this method are now sufficient to calculate the desired causal metrics as outlined in section 1.4. Details of these calculations are given in the post hoc analysis in part II.

Part II

Experiments and results

Chapter 3

Total effects and grammaticality

Using methods of causal mediation, we perform a number of experiments to investigate a number of effects on various components of the model. We begin with total effect, calculating the difference in probabilities for different verb inflections for prompts of different number. To further understand these effects, we further decompose the total effect to discover the source of the differences across syntactic structures.

3.1 Total Effect

In causal analysis, the total effect is defined as the effect of changing some independent variable X on some dependent variable Y. In our experiments, we are interested in how strongly models prefer correct verb forms. To measure this, we first identify modifications to the prompt fed to the model as the independent variable, and the output probabilities of a verb's number inflections as the dependent variable. We can then obtain the total effect by comparing the model prediction probabilities of verbs when preceded by a singular subject to the probabilities when preceded by a plural subject.

To formalize this, we need to define a measure for a model's preference for correct verb forms. Given a verb $v \in \mathbf{v}$ and prompt $u \in \mathbf{u}$ with number feature *sg*, we use the relative prediction probabilities of the grammatically correct v_{sg} and incorrect v_{pl} verb forms as a measure of the preference of a model with parameters θ for the grammatically correct inflection:

$$y(u, v) = \frac{p_\theta(v_{pl} \mid u_{sg})}{p_\theta(v_{sg} \mid u_{sg})} \tag{3.1}$$

where $y < 1$ indicates a preference for the correct inflection given a singular prompt, and $y > 1$ indicates a preference for the incorrect inflection.[1]

To obtain our independent variable, we define a class of interventions $\mathbf{x}$ that modify the prompts in $\mathbf{u}$ in a systematic way. As we are eventually concerned with the ability of models to choose correct inflections despite the presence of distractors and attractors, we define the intervention `swap-number`, which replaces the prompt subject with the same lexeme of the opposite number inflection (e.g., change "author" to "authors" or vice versa) u_{pl}. We also define the `null` intervention, which leaves u_{sg} as-is (following the methods of Vig et al. 2020).

Returning to our dependent variable, we now define $y_x(u, v)$ as the value of y under intervention x on prompt u. Because the intervention `swap-number` entails swapping the subject for a noun of the opposite number, we now expect $y_{\texttt{swap-number}} > 1$ in eq. (3.1) if the model behaves as expected, since the verb that was originally the correct inflection is now incorrect and vice versa.

Having established out independent and dependent variables, we now define the total effect (TE) for prompt u as the relative difference between the probability ratio y under the `swap-number` intervention and the ratio under the `null` intervention:

$$\begin{aligned}
\mathrm{TE}(\texttt{swap-number, null}; y, u, v) &= \frac{y_{\texttt{swap-number}}(u, v) - y_{\texttt{null}}(u, v)}{y_{\texttt{null}}(u, v)} \\
&= \frac{y_{\texttt{swap-number}}(u, v)}{y_{\texttt{null}}(u, v)} - 1 \\
&= \frac{p_\theta(v_{pl} \mid u_{pl})}{p_\theta(v_{sg} \mid u_{pl})} \Big/ \frac{p_\theta(v_{pl} \mid u_{sg})}{p_\theta(v_{sg} \mid u_{sg})} - 1
\end{aligned} \tag{3.2}$$

We interpret this quantity as the overall preference of a model for the correct inflection of v in context u. Since we observe both singular and plural versions of the prompt, it is arbitrary which grammatical number we start with.

In our analysis, we are interested in the average total effect across prompts and verbs:

$$\begin{aligned}
&\overline{\mathrm{TE}}(\texttt{swap-number, null}; y) = \\
&\mathbb{E}_{u,v}\left[\frac{y_{\texttt{swap-number}}(u, v)}{y_{\texttt{null}}(u, v)} - 1\right]
\end{aligned} \tag{3.3}$$

We calculate the average total effect for each syntactic construction for different sizes of GPT-2. Unlike in Linzen et al. [2016], we do not measure accuracies by checking whether one probability is higher than another. Rather, the total effect quantifies the margin between the probabilities of correct and incorrect continuations with some intervention.

[1] We arbitrarily choose to start with *sg*; we can swap *sg* and *pl* in eq. (3.1) without loss of generality since we do not directly observe y. This is clarified after eq. (3.2).

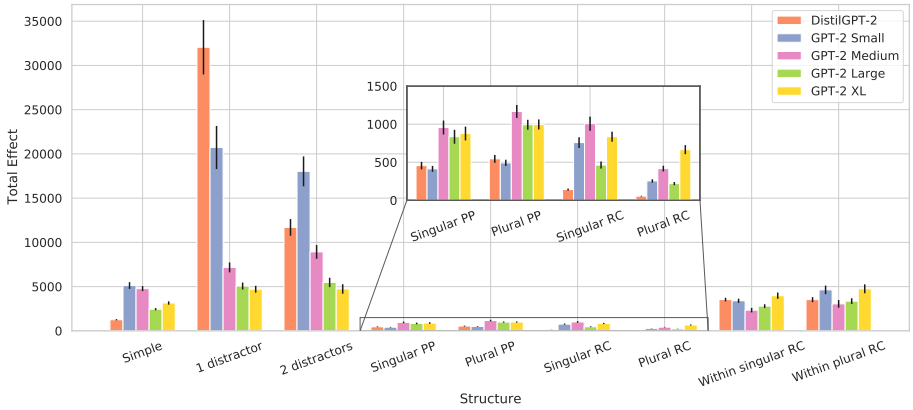

Figure 3.1: Total effects for each structure by model size for GPT-2. Adverbial distractors increase total effects, while attractor phrases decrease them.

Because larger models tend to exhibit correct subject-verb agreement more often than smaller models [Hu et al., 2020, van Schijndel et al., 2019], we hypothesize that larger models will generally have larger TEs for the same structure (i.e., we predict that higher accuracies are indicative of larger margins between probabilities).

3.1.1 Results

Figure 3.1 presents total effects by structure for various sizes of GPT-2. With randomized weights, TEs are always near-zero, as expected, and as such are not shown in the figure.

In 'simple agreement' and 'within RC' (where there is no separation of subject and verb, TEs vary between 1,000 and 5,000, depending on model size. This is far higher than the TEs below 250 reported for gender bias in Vig et al. [2020], which is to be expected: GPT-2's training objective explicitly optimizes for predicting grammatically correct tokens given context, but does not optimize for unbiased gender assignment. Analyzing our results, we arrive at a number of conclusions.[2]

Compared to small models, larger model do not have a stronger preference for correct inflections. Contrary to our hypothesis, and unlike Vig et al. [2020], we do not observe larger TEs for larger models, indicating that models do not necessarily prefer correct inflections more as models scale. While larger GPT-2 models have been shown to achieve higher accuracy on targeted syntactic evaluation sets [Hu et al., 2020], our results show that this performance advantage is not necessarily due to a larger preference for correct inflections.

[2]We discuss the (non-)impact of complementizers on the total effect in section 4.3.

Adverbial distractors increase correct inflection preference. TEs are highest for structures where distractors are present, with DistilGPT-2 and GPT-2 Small attaining the highest TEs in such contexts. This is surprising, as we expect subject-verb agreement accuracy to decline as the distance between the subject and the verb increases. We suspect the adverbs are acting as cues that a verb will soon appear, thus increasing the probability of both the correct and incorrect verb, but increasing that of the correct verb more. Additional analysis confirms that verb probabilities are notably higher in the distractor structures; see section 3.2.1.

Interposing attractors decrease correct inflection preference. When PPs or RCs separate the subject and verb, TEs decrease. The number of the attractor does not significantly change TEs across PPs, but does have a more notable effect across RCs: GPT-2 is more certain of its choices across singular RCs than across plural RCs, as evidenced by higher TEs for the former. This may point to a bias within the model, which will be discussed further in section 3.2.

Total Effects Across Architectures

Figure 3.2 presents total effects for all structures across architectures. The magnitude of total effect is generally similar for XLNet and GPT-2 (except when dealing with relative clauses), whereas total effects are much smaller for Transformer-XL. It seems that parametrization and model depth do not correlate well with total effects.

Perhaps the effects for Transformer-XL are smaller due to the longer effective contexts it has, which could make it prone to assigning smaller probabilities to a larger set of tokens than GPT-2 while still behaviorally performing well. However, it is harder to explain the similarity between GPT-2 and XLNet, given the great differences between them in training and the divergence in their behavior as revealed by the indirect effect results in section 4.1.2.

3.2 Grammaticality Margin

Noticing that the plurality of attractors makes a difference in total effects raises the question of whether models have equal preference for correct inflections when given singular and plural subjects. In other words, whether singular and singular agreement are treated equally by language models.

Because total effects do not disentangle model preferences attained when the subject is singular or plural, we also propose the grammaticality margin metric (shortened to "grammaticality" hereafter) to aid in the total effect analysis. This metric allows us to observe whether inflectional preferences are larger for singular or plural prompts, whereas total effects measure preferences without respect to grammatical number. Grammaticality is defined as the probability ratio of the correct inflection over the incorrect inflection given u with a specific number

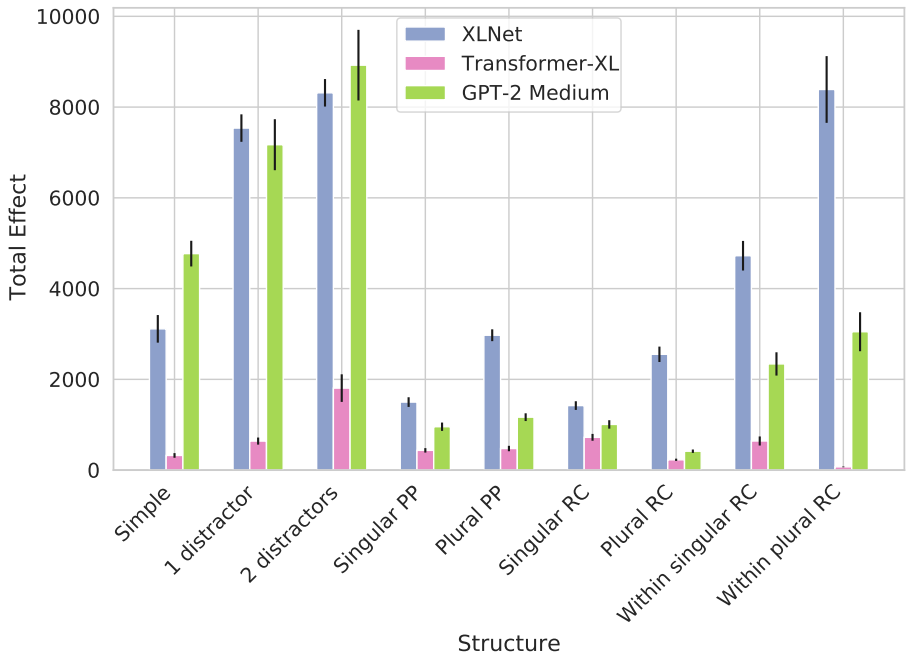

Figure 3.2: Total effects across structures by architecture (ordered by increasing number of layers/increasing parameterization).

		Tense	
		Singular	Plural
Person	First	*run*	*run*
	Second	*run*	*run*
	Third	***runs***	*run*

Table 3.1: English present-tense forms of the verb *to run*.

feature *sg* or *pl*:

$$G(u_{sg}, v) = \frac{p_\theta(v_{sg}|u_{sg})}{p_\theta(v_{pl}|u_{sg})} \tag{3.4}$$

This is equivalent to the reciprocal of $y_{\texttt{null}}$ in the u_{sg} case. Unlike total effects, whether u starts as singular or plural *does* matter here, since we are explicitly measuring the effect of grammatical number on inflectional preferences.

We expect grammaticality values to be higher when the subject and attractors match in number than when they do not match. We also expect grammaticality values to be similar for singular and plural subjects; however, differences can indicate biases toward a certain grammatical number in general.

3.2.1 Results

Throughout this section, we often refer to structures as "having" grammaticality values for convenience. A more accurate description would say that the output of the model has some grammaticality value when the input has a specific syntactic structure.

Models prefer plural verb forms. Figure 3.3 presents grammaticality values separately for singular and plural subjects, as well as singular and plural attractors when applicable. It is immediately apparent that plural subjects always have higher grammaticality values regardless of the structure or attractor number. In other words, GPT-2 forms stronger agreement dependencies between verbs and *plural* subjects than singular subjects.

We speculate that this phenomenon may be due to the nature of the number inflection in English verbs. In English, the present-tense third-person plural of a verb takes on an unmarked present tense form of the verb, while the third-person singular takes on a marked form. This is illustrated in table 3.1 for the verb *to run*. Since the unmarked form of the verb is often much more frequent, it makes sense that the language model may learn a preference for unmarked verbs during training time. This asymmetrical treatment of singular and plural agreement is discussed further at the end of this section.

Models are misled by attractors. Grammaticality increases when attractors and subjects match compared to when they do not match. This mirrors

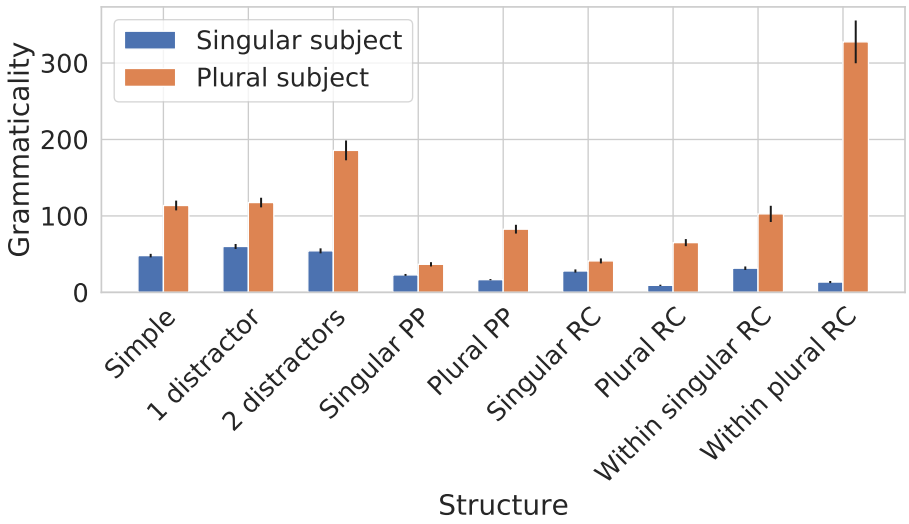

Figure 3.3: Grammaticality for each structure for GPT-2 Medium. The subject number (indicated by bar color) refers to the grammatical number of the subject with which the target verb agrees; the number in the structure name refers to the grammatical number of the attractor (in structures where attractors are present).

a pattern in humans who often mistakenly inflect verbs to agree with a more recent noun than the subject [Bock, 1995]. We expect to see an improvement in grammaticality when subjects and attractors match as there is no potentially conflicting evidence for the number of the verb. Although this effect is dominated by the disparity in plural and singular subject grammatically, the disparity in grammaticality is significantly less when the attractor is singular. This effect is more apparent when considering the relative sizes of bars of a single color in fig. 3.3. For plural subjects, plural attractors result in higher grammaticality, and similarly for singular subjects, singular attractors result in higher grammaticality. The same effect is seen both in across and withing RC cases, further supporting our hypothesis.

Models are sensitive to the syntactic structure of interposing tokens. This is evidenced by the contrast between interposing attractors which decrease grammaticality, regardless of plurality, and adverbial distractors which have little or the opposite effect, despite the fact that the 'across two distractors' structure places the same token distance between subject and verb as 'across a PP'. This means that distance between subject and verb is less important than the type of the structure separating them.

To investigate the cause of this effect, we further decompose the grammaticality metric into its component probabilities. Figure 3.4 shows that separating the

subject and verb with adverbs tends to increase the probability of the verb. Regardless of whether the subject and verb agree, adding adverbs does always increase the probability of verbs. Note the log scale: we observe visually similar increases in log probabilities after adding 1 or 2 adverb distractors, but visually similar differences at higher points in the graph are actually much larger increases. This supports our hypothesis from section 3.1.1 that adverbs increase all verb probabilities, but that they increase correct inflection probabilities more.

Subject-preceding attractors have subject number-dependent impacts on grammaticality. In the 'within singular RC' structure, grammaticality is reduced for both singular and plural subjects compared to the 'simple agreement' structure. However, 'within plural RC' has a polarizing effect: in the singular subject case, grammaticality is significantly reduced when subject and attractor do not match in number, but in the plural subject case, grammaticality is instead greatly increased when the subject and attractor do match.

This finding lends support to the hypothesis from that GPT-2 treats singular and plural agreement asymmetrically. Jumelet et al. [2019] report related findings that LSTM LM's have "default" verb forms for number agreement. In particular, they find that the singular verb form serves as the default, and additional evidence, such as a plural subject, is necessary to produce a plural verb prediction. This same type of asymmetrical and perhaps "default" preference for a verb form in Transformers may contribute to the patterns we see in our experiments.

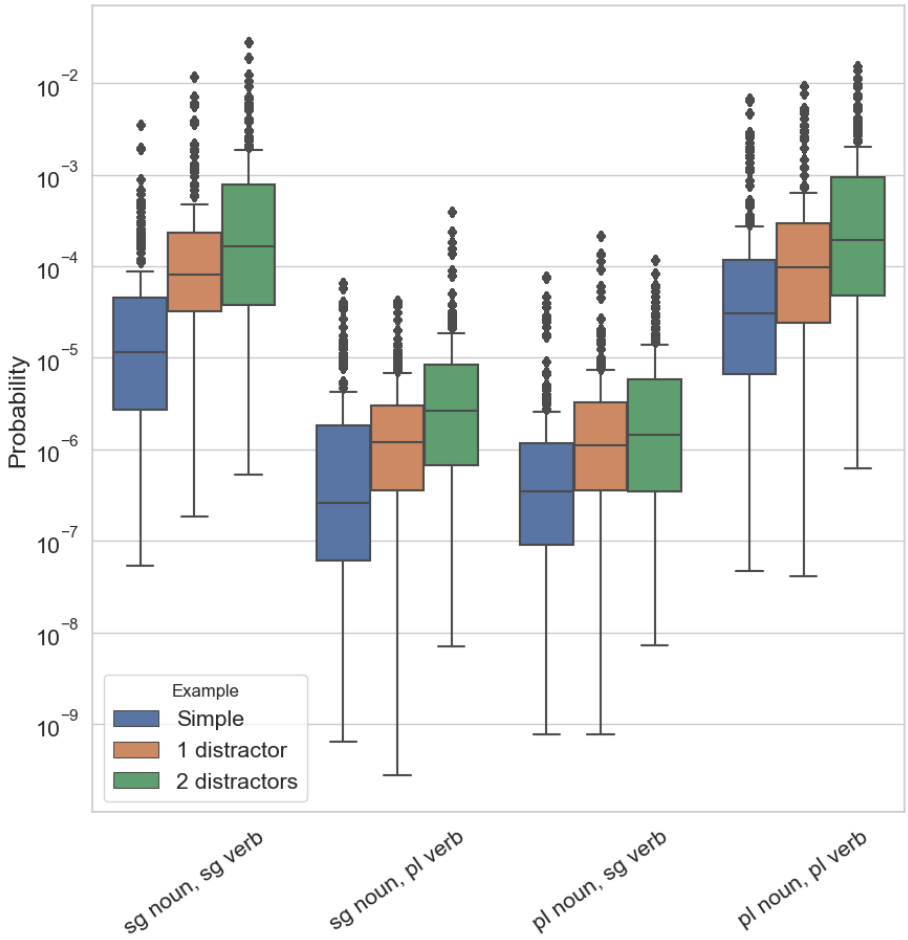

Figure 3.4: The distribution of target verb probabilities for 'simple agreement', 'across one distractor', and 'across two distractors'. Note that the y-axis uses a log scale.

Chapter 4

Natural Indirect Effect

Having investigated the total effects of changing the prompt subject number, we now turn to *indirect effect* in order to investigate the contributions of individual model components as mediators of syntactic number agreement. In our experiments, we are interested in determining which model components mediate number agreement by measuring the indirect effects of model components on the output of the model. We can find this, according to eq. (1.3), by allowing Y to be the output metric defined in eq. (3.1), setting X to the interventions `swap-number` and `null`[1], and setting Z to be the output of a component within the model.

The type of indirect effect we are interested in is called the *natural* indirect effect because we set Z to the value it would take in a counterfactual scenario, rather than setting it to a default value like 0. To formalize this, the natural indirect effect (NIE) is the change in effect when a prompt u is unchanged, but a model component $\mathbf{z}$ (e.g., a neuron) is set to the value it *would have taken* if the intervention had occurred.

$$\begin{aligned}&\overline{\text{NIE}}(\texttt{swap-number, null}; y, \mathbf{z}) = \\ &\mathbb{E}_{u,v}\left[\frac{y_{\texttt{null},\mathbf{z}_{\texttt{swap-number}}(u,v)}(u,v)}{y_{\texttt{null}}(u,v)} - 1\right]\end{aligned} \tag{4.1}$$

This allows us to evaluate the contribution of specific parts or regions of a model to the syntactic preferences we observe. More specifically, we can measure to what extent the total effect of swapping the subject on inflection preferences can be attributed to specific model components. Here, we independently analyze the individual neuron NIEs and discover, among other things, a bifurcated mechanism for performing number agreement. We also attempt to analyze

[1] Recall section 3.1

attention heads, though we find that they do not present consistent interpretable results.

4.1 Neuron indirect effects

Based on the findings of prior probing work on dependency parsing [Hewitt and Manning, 2019], we hypothesize that neuron NIEs will peak in the upper-middle layers for all models. Additionally, since GPT-2 and Transformer-XL always process input left-to-right, we expect differing indirect effect results between different structures, but similar indirect effect results overall across these two model architectures. Because XLNet is exposed to all word order permutations of its input sentences during training, we hypothesize that results may be more similar across syntactic structures, and that it will display different indirect effect trends compared to the other two models.

4.1.1 Neuron natural indirect effects in GPT-2

For each GPT-2 model size, we select the 5% of neurons with the highest NIE in each model layer; fig. 4.1 compares NIEs across model sizes, and fig. 4.2 compares NIEs across structures for GPT-2 Medium.[2] When weights are randomized, indirect effects peak at layer zero—albeit at values perhaps too small to be meaningful—and then remain close to zero in higher layers. This indicates that the vast majority of the indirect effect observed for trained models is due to what is learned, rather than chance.

GPT-2 have bifurcated number agreement mechanisms. We observe two distinct layer-wise patterns for GPT-2. In structures where the target verb directly follows the subject ('simple agreement' and 'within RC', the top 3 plots in fig. 4.1), NIEs continually increase in higher layers. Conversely, for structures with subject-verb separation ('across one/two distractor(s)', 'across PP', and 'across RC', the bottom 3 figures in fig. 4.1), NIEs peak at layer 0 and (more notably) in the upper-middle layers. The effect then decreases sharply, reaching zero in the uppermost layers.

Even a single token between subject and verb brings about this second indirect effect contour, indicating that distance is a less important factor than the presence of *any* separation in invoking this second syntactic agreement mechanism. These two distinct indirect effect contours may indicate two distinct subject-verb agreement mechanisms, depending on subject-verb separation. This resembles a finding of Lakretz et al. [2019], who found distinct systems for short- and long-distance agreement in LSTMs. The peak mean NIE is lower for cases with non-adjacent subjects and verbs than for structures where there is no separation,

[2] We also produced figures using *all* neurons. In these, the contour of the graph across layers does not change, but the magnitudes were lower since we average over more neurons.

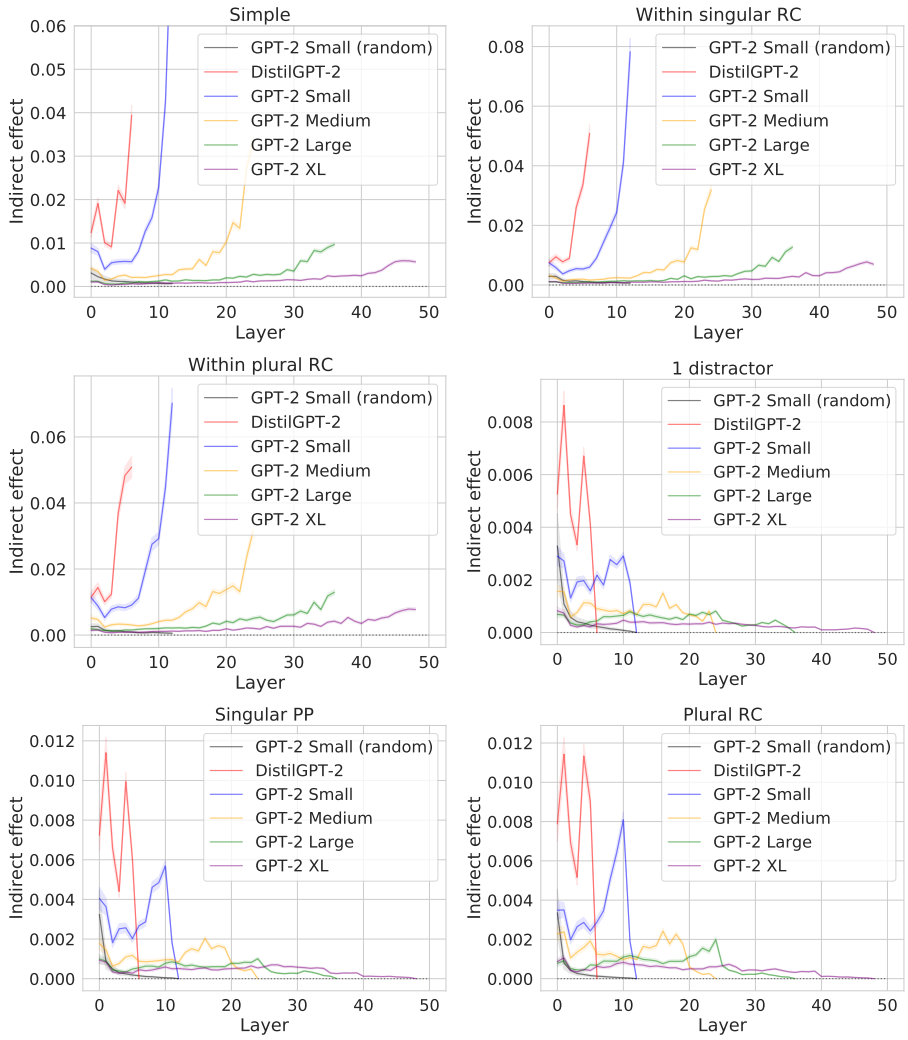

Figure 4.1: Natural indirect effects the top 5% of neurons in each layer of various GPT-2 sizes. Each figure focuses on a single structure and compares across GPT-2 sizes.

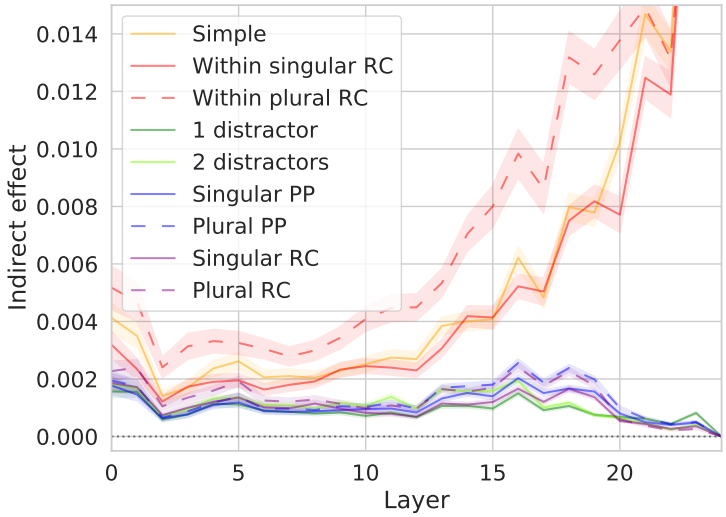

Figure 4.2: Natural indirect effects of the top 5% of neurons in each layer of GPT-2 Medium.

possibly indicating that syntactic agreement information is more distributed among neurons when separation occurs.

Structural knowledge is more distributed in large models, but agreement mechanisms are similar across model sizes. For each structure, the maximum NIE per layer is always lower for larger models. Peaks in NIEs are also more distributed across layers for larger models. This suggests that structural knowledge is more concentrated in fewer neurons with stronger inflectional preferences in smaller models, whereas it is more distributed across neurons in larger models. Nonetheless, the overall contour of NIEs is similar across model sizes for a given structure, indicating that the underlying mechanisms of agreement are similar across model sizes.

4.1.2 Comparison to Other Architectures

We also investigate the neuron NIEs of Transformer-XL (fig. 4.3) and XLNet (fig. 4.4) to observe whether syntax is represented in a similar manner across models (for total effects across architectures, see section 3.1.1).

We observe similar dichotomous agreement mechanisms across GPT-2 and Transformer-XL. 'Simple agreement' and 'within RC' show similar mechanisms of subject-verb agreement, whereas separating subject and verb lead to very different mechanisms of agreement. Moreover, the contour of the NIEs across layers is similar for these two architectures. This supports our hypothesis

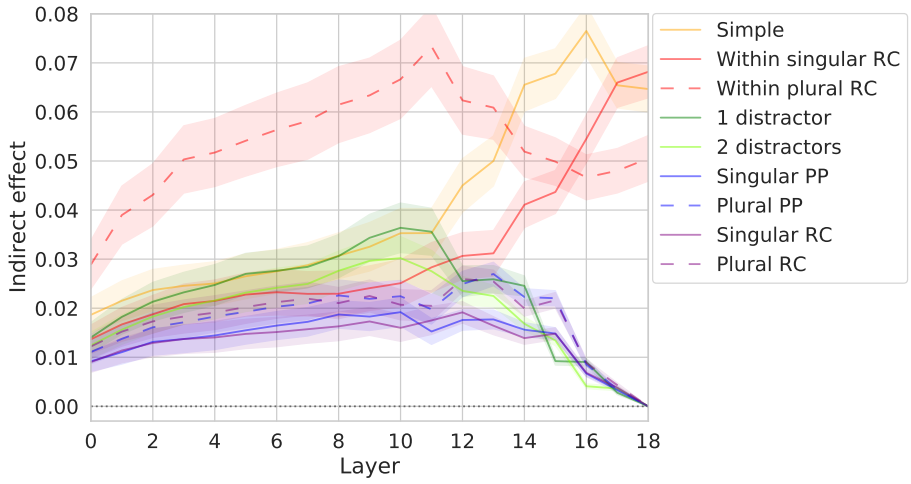

Figure 4.3: Natural indirect effects of the top 5% of neurons in each layer of Transformer-XL.

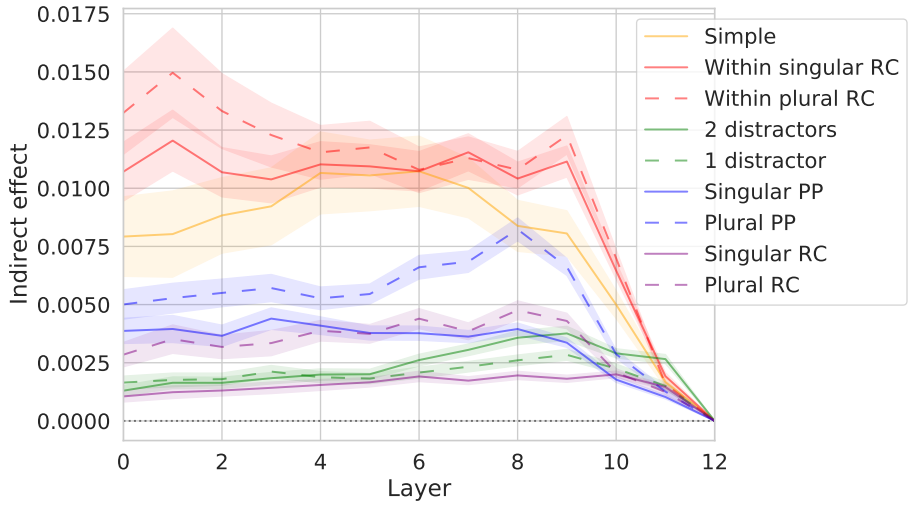

Figure 4.4: Natural indirect effects of the top 5% of neurons in each layer of XLNet.

that GPT-2 and Transformer-XL encode syntax in a similar manner.

Conversely, indirect effects in XLNet are different to those seen in GPT-2. We do not observe the same dichotomous behavior between subject-verb adjacent and subject-verb non-adjacent structures; rather, the overall contours are all the same. All of the indirect effects approach zero in the final layer. Overall, this resembles the way GPT-2 and Transformer-XL treat structures where subject and verb are not adjacent. This could be due to XLNet seeing all word order permutations of training sentences, thus preventing it from learning a dual heuristic for subject-verb number agreement. This supports our hypothesis that XLNet encodes syntax in a different manner to GPT-2.

4.1.3 Neuron Overlap Across Structures

To investigate whether the same neurons are implicated in subject-verb agreement across structures, we select the top 5% of neurons per layer by NIE and calculate the proportion of these high-NIE neurons that overlap between each pair of structures.

To investigate whether the extent of neuron sharing across structures correlate with human intuitions of syntactic similarity, we begin by hypothesizing syntactic similarities between structures based on the following linguistic features: distance between subject and verb; presence of adverbial distractors, a relative clause, prepositional phrase, and/or a noun attractor; and the number of the noun attractor when present. Appendix B details how we calculate similarity.

To quantify the similarity of the hypothesis matrix and a neuron overlap matrix, we calculate the L_1 norm of the element-wise difference between the lower-left triangle of both matrices, as the matrices are symmetric (using the L_2 norm does not change which layer has the lowest difference norm). We exclude the diagonal, which always contains 100% similarity/overlap.

For each model, we present neuron overlaps for the layer with the lowest difference norm to the hypothesis (fig. 4.5). The lowest difference norms are 443 (GPT-2), 510 (Transformer-XL), and 486 (XLNet). GPT-2 Medium's overlap across structures at layer 21 (of 24) is visually similar to the hypothesis, meaning that **this layer in GPT-2 shares neurons for subject-verb agreement across structures in a way that aligns with human intuitions about syntactic similarity.** Interestingly, it learns to do this without receiving explicit syntactic supervision during training.

Layer 15 (of 18) of Transformer-XL displays similar trends to GPT-2, though the extent of overlap is higher across structures in general here. There is more significant overlap between the adverbial distractor structures and the structures that contain attractors. 'Simple agreement' also has more overlap with structures containing attractors than 'within RC', which is contrary to the hypothesis. Also note how 'across singular RC' has more overlap with 'across PP' than 'across plural RC' (and vice versa for 'across plural RC'), indicating that **the number**

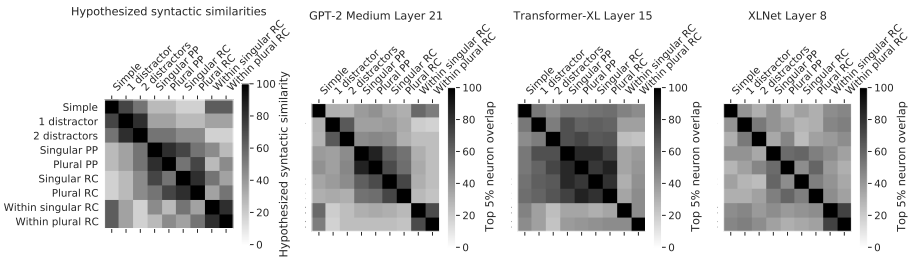

Figure 4.5: Hypothesized syntactic similarity across structures (left), as well as the overlap of the top 5% of neurons per-structure by indirect effect for GPT-2 (center-left), Transformer-XL (center-right), and XLNet (right).

of the attractor is more salient to Transformer-XL than the structure of the phrase containing the attractor.

Meanwhile, Layer 8 (of 12) of XLNet appears noisier. There is slightly more overlap between structures across noun attractors, but the extent of overlap is smaller compared to other models, suggesting that there are more neurons specialized to specific structures. However, the indirect effect findings for XLNet suggest a more unified mechanism for syntactic agreement across all structures; if this were the case, we would expect neuron overlap to be high, and for the extent of overlap to be similar across *all* structures, rather than being higher between more similar structures. We observe the latter, but not the former. Regardless, both observations further support our hypothesis that XLNet uses different mechanisms to encode syntax than the other two architectures.

Figure 4.6 presents neuron overlaps across all layers of DistilGPT-2. We first note that the overall extent of neuron overlap across structures tends to increase up to the upper-middle layers, before sharply decreasing in the highest layer to near-zero values. We find that this trend holds for all other sizes of GPT-2, as well as Transformer-XL; generally, overlaps continue to increase until the upper-middle layers, decreasing slightly in the second-highest layer and decreasing sharply to zero in the highest layer. Layer-by-layer difference (L_1) norms are presented in table 4.1.

4.2 Attention head indirect effects

4.2.1 Attention heads do not significantly mediate number agreement.

There are few major trends in attention head indirect effects. Each structure appears to implicate different heads in model behavior; even changing the number of the attractor has a large effect on which heads exhibit large NIEs. We present the mean indirect effects across prompts of each attention head for a sample of

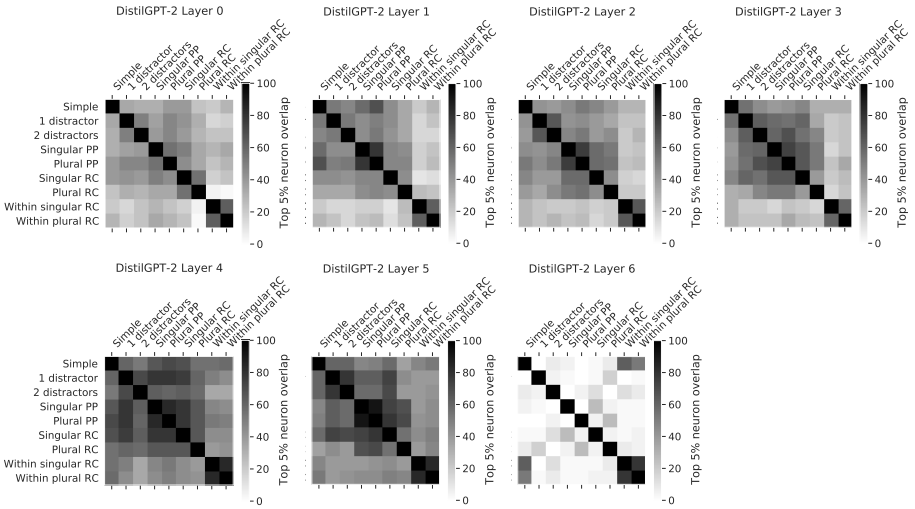

Figure 4.6: Overlap between structures of the top 5% of neurons in each of layer of DistilGPT-2 by indirect effect.

Layer No.	Diff. Norm
0	677
1	652
2	565
3	583

Layer No.	Diff. Norm
4	627
5	510
6	1301

Table 4.1: Difference L_1 norms between the hypothesis matrix and each layer of DistilGPT-2.

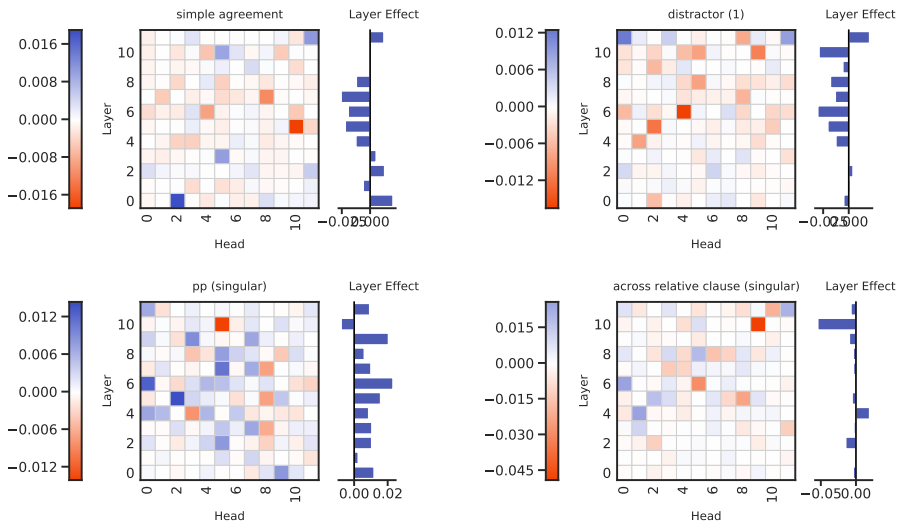

Figure 4.7: Attention indirect effects for GPT-2 Small under the `swap-number` intervention.

structures for GPT-2 Small, under the `swap-number` intervention in fig. 4.7.

The trends we do see are that attention heads in the upper-middle layers seem to account for most of the positive *and* negative indirect effects, and that head 10-9 seems to have negative indirect effects for most structures where there is a separation between subject and verb, except 'across plural RC'. However, we do not observe strong indirect effects for head 10-9 when subject and verb are adjacent. Head 11-11 has the most consistently positive indirect effects across structures, though its magnitude is typically low.

Indirect effects are largely positive for 'within plural RC', but otherwise, indirect effects are relatively evenly split between positive and negative. Indeed, the sum of indirect effects across heads for most structures is very close to 0, with many sums being a low-magnitude negative number. This indicates that these attention indirect effects may simply be noise.

Our best explanation is that most of the total effect does not flow through the attention heads, making the model output robust to attention head intervention. This is supported by our finding that the sum of the indirect effects across heads averaged across prompts is always near-zero (and is often small and negative), as is the indirect effect attained when intervening on all attention heads simultaneously. Meanwhile, the direct effect across heads is very large and captures most of the total effect. See fig. 4.8. This indicates that most of the total effect does not flow through the attention heads, but through other model components (e.g., word representations, neurons). Thus, perhaps what specific heads are displaying across structures is noise, rather than meaningful effects.

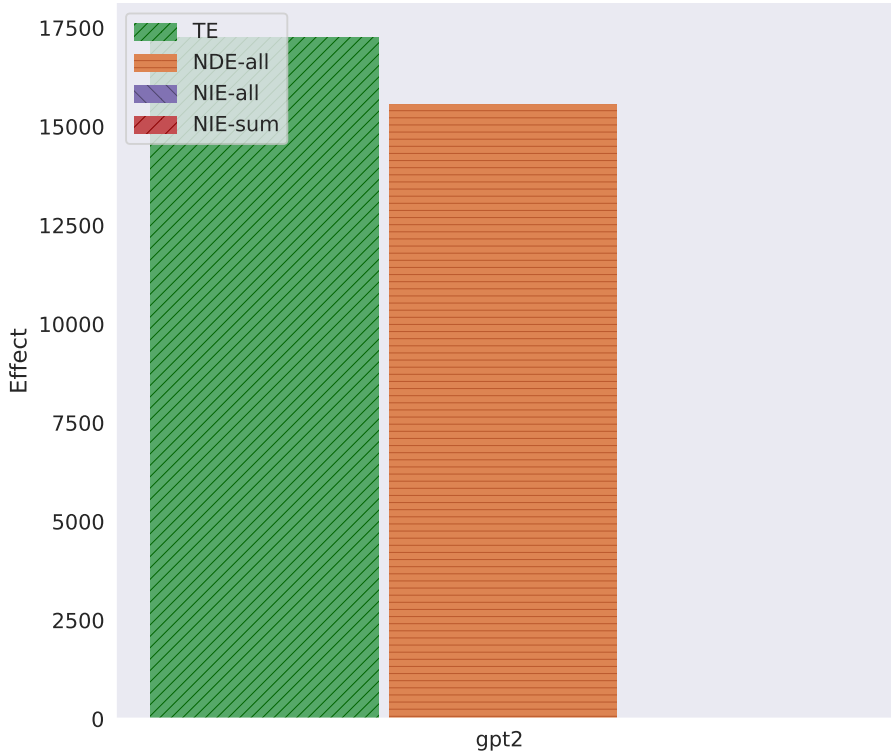

Figure 4.8: Total (TE), direct (NDE-all), and indirect effects (NIE-all and NIE-sum) for attention heads for the 'across a distractor' structure. NIE-all (the indirect effect from simultaneously intervening on all attention heads) and NIE-sum (the indirect effect from summing independent and separate interventions on each attention head) are near zero and thus not visible.

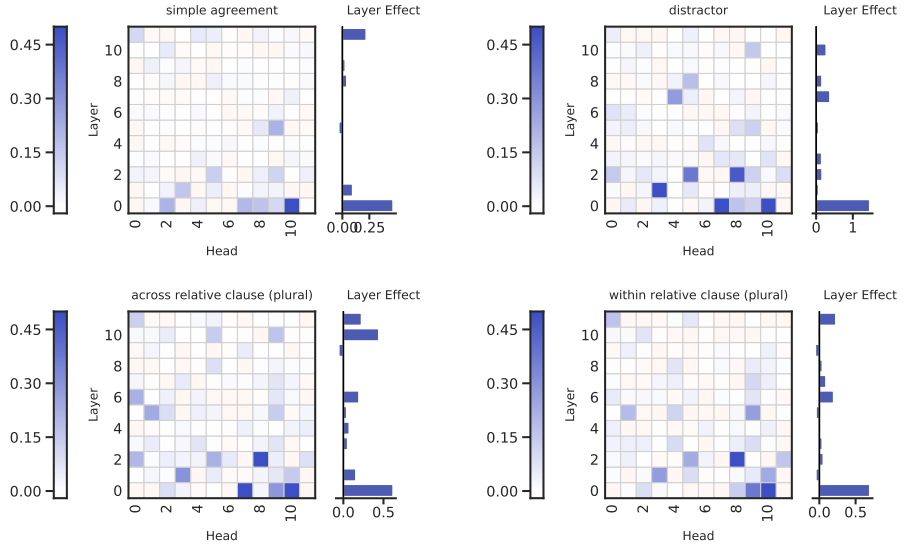

Figure 4.9: Attention indirect effects for GPT-2 Small under the `zero` intervention.

4.2.2 Controlled indirect effect

To test whether the model is simply robust to the `swap-number` intervention on attention heads, we define an alternate intervention `zero`: rather than intervening using the head output from swapping the number of the subject, we set a specific attention head **z**'s value to 0 and observe how this changes the effect on the unchanged prompt. This may be interpreted as the controlled indirect effect, as defined in Pearl [2001]. Figure 4.9 gives us the results for this approach.

For this intervention, trends are much more consistent across structures. Head 0-10 is always strongly implicated in behavior; since this is in the bottom layer, this indicates that much of attention's contribution to syntax is lexicalized. In other words, attention's contribution to subject-verb agreement may largely be based on which word embeddings are present in the input, rather than any compositional structural information that would be present in the upper layers. Qualitative analysis reveals that head 0-10 and head 2-8 always focus on the 2nd and 5th words in the prompt, respectively. Thus, attention's contribution to subject-verb agreement may largely be based on where important tokens appear in the input, rather than any abstract structural information that would be composed in the upper layers.

For all structures except where we have adverbial distractors, we see consistent positive indirect effects in the uppermost layers. No single attention head appears strongly implicated (though head 11-0 is always positive), but the layer effect is consistently positive and sometimes nears the magnitude of the layer effect

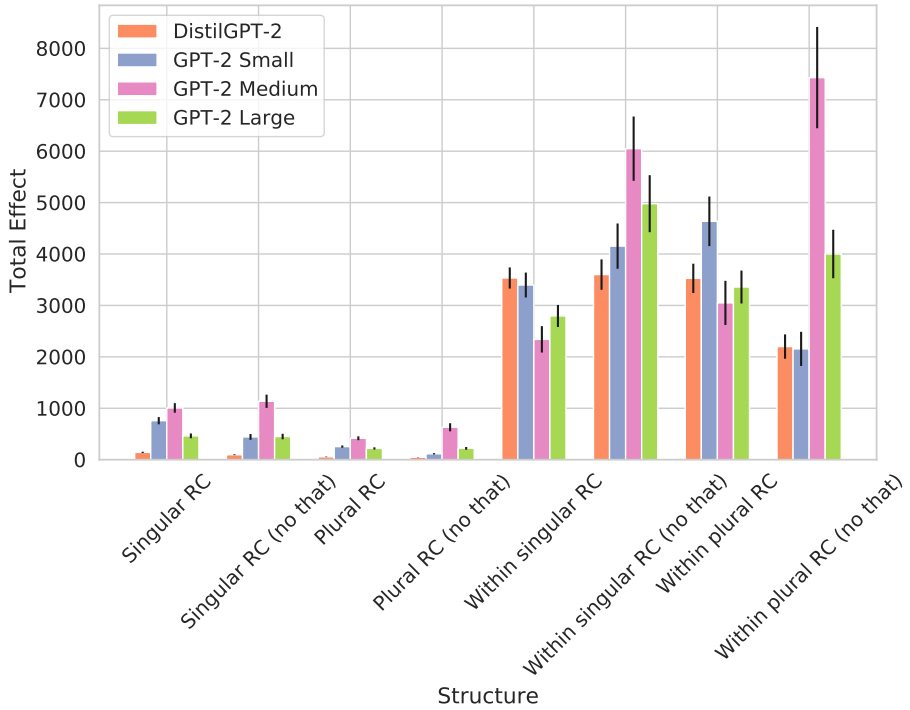

Figure 4.10: Total effects for 'across relative clause' and 'within relative clause' structures, with and without the complementizer *that*.

of layer 0. This indicates that more abstract structural information is quite distributed across attention heads in the uppermost layers, but nonetheless may be present.

4.3 The (Non-)Impact of Complementizers

We investigate the effect of including or excluding the complementizer *that* for the 'across RC' and 'within RC' structures, observing both TEs (fig. 4.10) and neuron indirect effects (fig. 4.11). While we expect lower TEs when the complementizer is absent, we observe only minor reductions in total effects in 'across RC'; this holds across model sizes. For 'within RC', however, trends are size-dependent. DistilGPT-2, GPT-2 Small, and GPT-2 Large appear mostly robust to the presence or absence of the complementizer, though GPT-2 Small does have lower total effects in the 'across plural RC' structure when *that* is absent. Meanwhile, GPT-2 Medium more strongly prefers correct inflections when *that* is absent. It is not immediately clear why this is the case, because deleting the complementizer introduces more ambiguity.

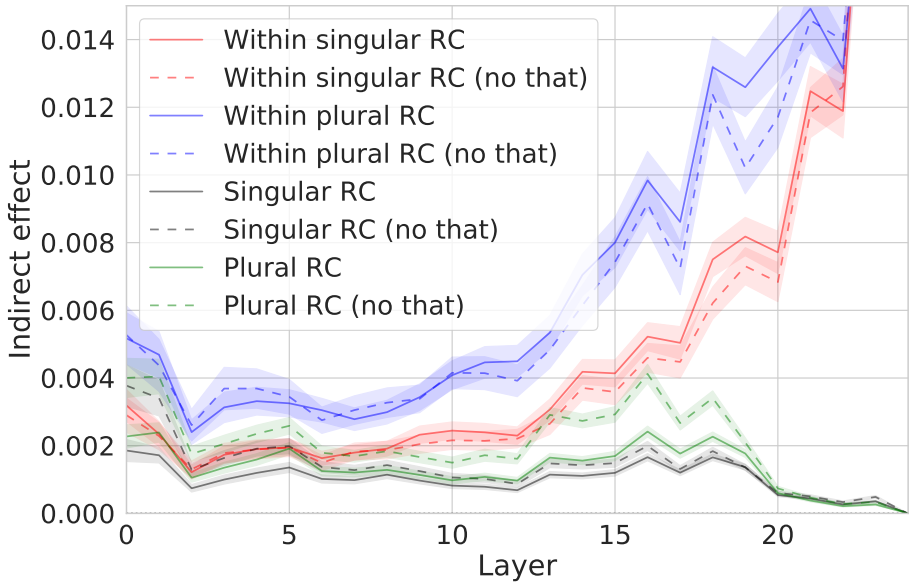

Figure 4.11: Indirect effects for 'across relative clause' and 'within relative clause' structures, with (solid lines) and without (dashed lines) the complementizer *that*.

One might expect that the exclusion of the complementizer will change neuron behavior when agreeing subjects and verbs. However, there does not appear to be any significant difference in magnitude or contour of the indirect effects across layers when including or excluding it. Thus, while excluding the complementizer can make subject-verb agreement slightly more difficult for LMs [Marvin and Linzen, 2018], it does not appear to change the mechanisms through which subject-verb agreement happens in the model.

Chapter 5

Conclusion

In this thesis, we contribute a novel method and a number of results to the body of research investigating Transformer-based language models as psycholinguistic subjects. In particular, we provide a method to reveal and analyze the internal mechanisms that give rise to syntactic number agreement behavior in these language models in controlled settings. Using this method we locate within the model specific neurons and attention heads that contribute to number prediction and gain a better understanding of the internal mechanisms of transformer models. Among our more surprising findings, we find that larger models do not necessarily learn stronger preferences for correct inflections, and that that GPT-2 and Transformer-XL use two separate mechanisms to deal with number agreement, and the use of these mechanisms is dependent on whether the subject is adjacent to the verb.

Though our investigations cover a number of metrics and analyses, there are many questions that remain open for future work. For instance, we analyze interventions on individual neurons and attention heads and observe relatively small effects for most. As a next step, we recommend that future work investigate groups of model components, (i.e. groups of neurons) as mediating variables. This approach is used in Vig et al. [2020] for gender bias. We also recommend that future work investigate other syntactic phenomena such as filler-gap dependencies and negative polarity items.

Though our analysis is focused on Transformer-based language models, our approach is general enough to be applicable to a number of neural network architectures performing a number of tasks, even tasks outside the realm of language modeling.

Appendix A

Grammar

Tables A.1 and A.2 give the vocabulary used to generate the syntactic structures.

Singular	Plural
athlete	athletes
aunt	aunts
boy	boys
carpenter	carpenters
doctor	doctors
farmer	farmers
father	fathers
friend	friends
girl	girls
guy	guys
kid	kids
lawyer	lawyers
man	men
mother	mothers
poet	poets
singer	singers
teacher	teachers
uncle	uncles
victim	victims
woman	women

(a) Subjects

Singular	Plural
approves	approve
confuses	confuse
stimulates	stimulate
criticizes	criticize
understands	understand
avoids	avoid
greets	greet
admires	admire
discourages	discourage
inspires	inspire
knows	know
engages	engage
encourages	encourage
remembers	remember
observes	observe

(b) Verbs

Table A.1: Subjects and verbs

Singular	Plural
guard	guards
chef	chefs
architect	architects
skater	skaters
dancer	dancers
minister	ministers
driver	drivers
assistant	assistants
executive	executives
parent	parents

(a) RC nouns

Singular	Plural
praises	praise
calls	call
describes	describe
follows	follow
influences	influence
forgives	forgive
protects	protect
informs	inform
visits	visit
likes	like
hates	hate

(b) RC verbs

Singular	Plural
bike	bikes
car	cars
cat	cats
chair	chairs
desk	desks
dog	dogs
table	tables
tree	trees
truck	trucks
window	windows

(c) PP nouns

behind
beside
near

(d) PP prepositions

gently
openly
overtly
carefully
deliberately
knowingly

(e) Adverbs

Table A.2: RC, PP, and distractor structure vocabulary

Appendix B

Hypothesizing Syntactic Similarity

To generate the hypothesis similarity matrix between structures, we choose a set of features given in table B.1 that capture important syntactic information according to our intuition. Most of the features are binary; however, we also include a ternary feature and a numerical feature. The ternary feature, "attractor number", can take on values SG, PL, and 0 when there is no attractor.

To compute the similarity of two structures, we first sum the difference for each feature. For the binary and ternary features, the difference is 0 if the features have the same value, otherwise 1. For the numerical feature, we take the absolute value of the difference between distances, scaled to a value between 0 and 2. We scale the similarity to reduce the impact of the numerical feature on the total similarity.[1] Finally, we take the maximum possible difference across all pairs of structures, and subtract each pairwise distance from the maximum to obtain similarity scores. We normalize the similarities to the range [0, 100] by dividing similarities by the maximum possible similarity score; this is to make them more comparable to the neuron overlap matrices.

[1] We initially scaled this to be within the range [0, 1] like the other features, but this caused "within relative clause" to have high similarity to "across PP" and "across a relative clause". Thus, we increase its impact to better match our intuition.

Feature	Type
Subject and verb separated	binary
Tokens between subject, verb	numerical
Has adverbial distractor(s)	binary
Has noun attractor	binary
Attractor number	ternary
Has relative clause	binary
Has prepositional phrase	binary

Table B.1: Features (and their types) used in calculating hypothesized syntactic similarity.

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